LUMBERBANK

Midfloor and Rafter Estimation Tool

Phase 1 — Design Brief and Scope

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Author	Steve Penney, Technical Team Leader
Organisation	Lumberbank
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1. Purpose and Scope

This document defines the design brief and functional scope for Phase 1 of the Lumberbank Midfloor and Rafter Estimation Tool. It is intended to provide sufficient detail for internal review and, where required, external development engagement.

Phase 1 delivers a web-based estimation tool enabling Lumberbank's merchant and pre-nailer customers to generate indicative cost estimates for midfloor and rafter timber systems using Lumberbank products. The tool captures job and estimate data to support future analysis and forms the foundation for Phase 2, which will introduce a full structural design engine.

2. Background and Strategic Context

Lumberbank operates in a competitive structural timber wholesale market where pricing is largely aligned across suppliers. With new competitors entering the region, the business must develop a clear value proposition beyond price.

A key strategic insight is that Lumberbank's primary competitor has embedded their design tool into engineers' and specifiers' workflows, meaning their products are frequently named at the design stage. Products specified on consented drawings are rarely substituted downstream, creating a strong lock-in effect. Lumberbank must replicate this flywheel.

Additionally, the current free layout generation and quoting service lacks a direct link from design output to confirmed sales order, making conversion tracking difficult and cost recovery inconsistent. Phase 1 addresses this by creating a digital job record from first contact through to order.

This document covers Phase 1 only. Phase 2 (full structural design engine) and the PyRevit plugin integration are referenced where relevant but are explicitly out of scope for this phase.

3. System Overview

The tool is a web-based application built on a Flask/MySQL backend with a browser-based front end. It is hosted as a web application and accessible to authorised users via login.

The central object in the system is a Job. A Job represents a construction project and contains all associated estimates and uploaded documents. Jobs are created by merchant users and progress through a defined lifecycle.

3.1 Job Lifecycle

- Job created with project details by a merchant user
- One or more sub-jobs added, each representing a distinct floor or roof system to be estimated
- Estimate parameters entered per sub-job; cost estimate calculated and PDF generated
- If a detailed layout is required, drawing files are uploaded against the job
- Drawing upload triggers an automated email notification to the Lumberbank detailing team
- Lumberbank detailing team manages the full layout through their existing workflow

3.2 Architecture Summary

- Backend: Flask (Python), MySQL database
- Frontend: Browser-based web application
- PDF generation: Server-side, Lumberbank-branded template
- Email notifications: Automated on drawing upload trigger
- Hosting: Web application (platform TBC)

4. User Roles and Tenancy Model

The system is multi-tenant, organised around Organisations. Each Organisation represents a merchant or pre-nailer business. Users belong to an Organisation and inherit its access rights.

4.1 Organisation Types

Organisation Type	Description	Pricing Access
Merchant / Pre-nailer	External customer organisations. Primary users of the estimation tool.	Yes — wholesale prices visible
Designer / Specifier	Future user type for Phase 2 design tool. Not in scope for Phase 1.	No

Organisation type is controlled by an isMerchant flag at the organisation level. This flag gates access to pricing information and the estimation functionality. Non-merchant organisations will be introduced in Phase 2 and will access design features only.

4.2 User Roles

Role	Scope	Capabilities
Merchant User	Organisation-level	Create and manage jobs, run estimates, generate PDFs, upload drawings
Lumberbank Admin	System-level	Manage organisations and users, maintain price books, configure system settings (freight rules, surcharge), view all jobs
Lumberbank Detailing Team	System-level	Receive drawing upload notifications, view job details (Phase 1 only)

4.3 Onboarding and Access Control

Phase 1 onboarding is manual and controlled. The process is as follows:

1. Merchant submits an access request (mechanism TBC — email or web form)
2. Lumberbank admin reviews and approves or denies the request
3. Approved organisations are created in the system by Lumberbank admin
4. Users are pre-loaded by Lumberbank admin prior to merchant onboarding
5. Lumberbank demonstrates the tool to the merchant in person

The data model must accommodate self-service sign-up in future phases without requiring structural changes.

5. Job and Sub-Job Structure

5.1 Job

A Job is the top-level record representing a construction project. Each job belongs to a single Organisation (merchant).

Field	Description	Notes
Job Reference	User-defined reference or name	Required
Client Name	Name of the merchant's client	Required
Site Address	Physical address of the project	Required
Merchant	Auto-populated from the logged-in user's organisation	Read-only
Created Date	System-generated timestamp	Automatic
Status	Current lifecycle stage (e.g. Estimate, Drawing Uploaded, Order Placed)	System-managed

5.2 Sub-Job

A Job may contain one or more Sub-Jobs. Each Sub-Job represents a discrete floor or roof system within the project requiring a separate estimate. Examples include Unit 1 Midfloor, Unit 2 Midfloor, Unit 1 Roof, Unit 2 Roof.

Field	Description	Notes
Sub-Job Label	User-defined name (e.g. 'Unit 1 Midfloor')	Required
System Type	Midfloor or Roof	Dropdown
Estimate Data	All input parameters for the estimate	See Section 6
Calculated Sub-total	Wholesale cost for this sub-job	System-calculated
Member Schedule	Background member list used to build the price	Stored in DB, not shown to user

6. Functional Requirements — Estimation Module

6.1 Estimate Inputs

Estimate inputs are entered at the Sub-Job level. The input set is designed to be minimal while producing a reliable indicative cost. The following inputs are required per sub-job.

6.1.1 Floor / Roof Areas (Repeatable)

One or more areas may be defined within a sub-job to accommodate different joist types or spacings within the same floor or roof system.

Input	Type	Options / Notes
Area Label	Text (optional)	e.g. 'Bathroom', 'Main Floor'
Floor / Roof Area	Numeric (m²)	User-entered
Joist / Rafter Type	Dropdown	I-Joist, LVL
Joist / Rafter Size	Dropdown	Filtered by Joist Type. Product list maintained by LB Admin
Joist / Rafter Spacing	Dropdown	400 c/c, 450 c/c, 600 c/c

6.1.2 Perimeter Boundary Joists (Floor Sub-Jobs Only)

Double boundary joists are applied around the perimeter of the floor system. The perimeter is entered as a linear metre value either manually or, where derivable from area inputs, system-assisted. Boundary joists can be excluded from the estimate using a checkbox toggle.

Input	Type	Notes
Include Boundary Joists	Toggle (Yes / No)	Can be excluded for merchants where desired.
Perimeter (lineal m)	Numeric	Manual entry. System-assisted derivation to be evaluated during development.
Member Type / Size	Dropdown	LVL product. Filtered from product list.

6.1.3 Stair Void Trimmers (Floor Sub-Jobs Only)

A toggle input allows the user to include a standard stair void trimmer allowance. It is assumed that most midfloor systems will include one stair void. The tool applies a fixed lineal metre allowance (to be determined from historical job data) when the toggle is activated.

Input	Type	Notes
Include Stair Void Trimmers	Toggle (Yes / No)	One void assumed. Fixed lineal metre allowance applied when toggled on.
Member Type / Size	Dropdown	LVL trimmer product. Filtered from product list.

6.1.4 Additional Beams (Repeatable, Optional)

Allows the user to include any additional LVL or Glulam beams that are known at the time of estimation.

Input	Type	Notes
Member Type	Dropdown	LVL, Glulam
Member Size	Dropdown	Filtered by Member Type
Length (m)	Numeric	User-entered
Quantity	Numeric	User-entered

6.1.5 Roof Pitch (Roof Sub-Jobs Only)

When a sub-job is classified as a roof system, an additional input is required to account for the effect of pitch on rafter length.

Input	Type	Notes
Roof Pitch	Dropdown	Common pitch values (degrees or ratio). List TBC. A pitch factor is applied to convert plan area to rafter length.

6.2 Calculation Engine

The estimation engine derives a lineal metre quantity for each element and applies a unit price from the merchant's price book to produce a cost.

Element	Calculation Method
Floor joists	Lineal metres = Floor area (m²) ÷ Joist spacing (m)
Roof rafters	Lineal metres = (Roof area ÷ Joist spacing) × Pitch factor
Boundary joists	Lineal metres = Perimeter × 1.5 (double boundary parallel and single rim board perpendicular to joists)
Stair void trimmers	Fixed lineal metre allowance (to be calibrated from historical data)
Additional beams	Lineal metres = Length × Quantity

Unit prices are looked up from the merchant's price book by product. The total cost per sub-job is the sum of all element costs. The overall job total is the sum of all sub-job totals, plus applicable freight charges.

6.3 Price Books

Each merchant organisation has an associated price book maintained by Lumberbank Admin. Price books are stored in the database and contain wholesale unit prices (per lineal metre) for each product.

- Price books are managed by Lumberbank Admin only
- Prices can vary by merchant organisation
- Product list (joist types, sizes, beam types) is maintained by Lumberbank Admin and is shared across all price books
- Price book updates take effect on new estimates only

6.4 Freight Rules

Freight charges are calculated at the job level and applied to the total of all sub-jobs. Freight logic is configured by Lumberbank Admin and is not visible or adjustable by merchant users.

Condition	Charge Applied
Job total < Freight threshold (default $1,000)	Fixed freight fee applied
Job total ≥ Freight threshold	FIS (free into store). Freight surcharge applied if surcharge toggle is ON.
Freight surcharge toggle OFF	No surcharge shown. FIS only.

The following freight parameters are configurable by Lumberbank Admin:

- Freight threshold (dollar value, default $1,000)
- Fixed freight fee (dollar value, applied to sub-threshold orders)
- Freight surcharge toggle (on / off, system-wide)
- Freight surcharge percentage (applied to job total for qualifying orders)

7. Data Capture and Privacy

7.1 Data Captured

The following data is captured and stored against each job record:

- Job details: client name, site address, job reference, merchant organisation, date created
- Sub-job details: label, system type, all input parameters
- Background member schedule: lineal metre quantities and product codes per element per sub-job
- Calculated sub-totals and job total (wholesale cost)
- Freight charge applied
- Uploaded drawing files (filenames, upload timestamps)

The background member schedule is stored for Lumberbank's internal use only. It is not displayed to merchant users on the front end or included in the PDF output.

7.2 Data Use

Captured data will be used by Lumberbank for the following purposes:

- Sense-checking estimate accuracy against completed orders
- Calibrating m² rates and stair void allowances over time
- Informing sales and marketing activity (conversion tracking, merchant engagement)

- Identifying jobs that have progressed to order

7.3 Privacy

All data capture and storage must comply with Lumberbank's privacy policy and applicable New Zealand privacy legislation. Client and project data is associated with the merchant organisation and is accessible only to that organisation's users and Lumberbank Admin.

8. PDF Output Specification

8.1 Overview

The primary output of the estimation tool is a Lumberbank-branded PDF estimate document. The PDF is generated server-side and is available for download by the merchant user.

8.2 Addressee

The estimate is always addressed to the merchant organisation. It is intended for the merchant's internal use as part of their own quoting process to their client. The merchant is Lumberbank's customer.

8.3 Document Structure

The PDF estimate document shall contain the following sections in order:

Section	Content
Header	Lumberbank branding, document title, date
Project Details	Job reference, client name, site address, merchant name, estimate date
System Descriptions	Auto-generated plain-English description of each sub-job system, derived from input parameters. One paragraph per sub-job. Not editable by the user.
Itemised Estimate	One line per sub-job with label and sub-total price
Freight Charge	Freight fee or surcharge line (shown only if applicable)
Total	Total estimated cost (wholesale, ex GST)
Footer	Lumberbank contact details, disclaimer text, document reference

8.4 Auto-Generated System Description

The system description for each sub-job is composed automatically from the input parameters. It describes the specified members in plain English. Example:

> *"Unit 1 Midfloor — LIB240.88 I-joists at 600 c/c, LVL8 double boundary joists, stair void trimmers included, additional beams as specified."*

The user cannot directly edit the description text. If the description is incorrect, the user must return to the input screen and amend the relevant parameters.

8.5 Pricing Display

The PDF displays a single price per sub-job and a job total. Individual member quantities and unit rates are not shown on the PDF. The background member schedule is for internal use only.

All prices are wholesale costs (Lumberbank to merchant), exclusive of GST. A disclaimer to this effect shall appear on the document.

9. Notifications and Workflow Handoff

9.1 Drawing Upload Notification

When a merchant user uploads drawing files against a job, the system shall send an automated email notification to the Lumberbank detailing team.

Parameter	Detail
Trigger	Drawing file upload against a job
Recipient	quotes@lumberbank.co.nz
Email Content	Job reference, merchant name, site address, number of files uploaded, link or reference to access the job
Attachments	TBC — either files attached directly or accessible via the system

The Lumberbank detailing team will manage the job through their existing workflow upon receipt of this notification. No further in-system workflow is required for Phase 1.

10. Admin Controls

The following system configuration is managed by Lumberbank Admin only via an admin interface:

Setting	Description
Organisation management	Create, edit, activate, deactivate merchant organisations. Set isMerchant flag.
User management	Create, edit, deactivate users. Assign to organisations.
Product list	Maintain joist types, joist sizes, beam types available in dropdowns

Price books	Set and update per-lineal-metre wholesale prices per product per merchant organisation
Freight threshold	Dollar value below which fixed freight fee applies (default $1,000)
Fixed freight fee	Dollar value applied to sub-threshold orders
Freight surcharge toggle	System-wide on/off switch for the fuel surcharge
Freight surcharge percentage	Percentage applied to qualifying job totals when surcharge is active
Stair void trimmer allowance	Standard lineal metre allowance applied when stair void toggle is on

11. Out of Scope — Phase 1

The following items are explicitly out of scope for Phase 1 and are deferred to Phase 2 or later:

- Structural design engine (code-compliant member sizing to NZS1170)
- 3D interactive building frame model for member selection
- Detailed engineering calculation output
- Producer statements
- PyRevit / Revit addin integration
- Self-service merchant sign-up (manual onboarding only in Phase 1)
- Designer / specifier user type and associated access model
- Layout generation and full detailing workflow within the tool
- Direct integration with sales order system
- Conversion tracking dashboard (data is captured but reporting is deferred)

12. Open Questions and Decisions Required

#	Question	Owner	Priority
1	What is the standard stair void trimmer allowance (lineal metres)? To be calibrated from historical job data.	Technical Team	High
2	What are the common roof pitch values to include in the dropdown?	Technical Team	High
3	Confirm method for perimeter derivation — manual entry only, or system-assisted from area inputs?	Technical Team	Medium
4	Confirm initial product list for joist types, sizes, and beam types.	Technical Team	High
5	Confirm fixed freight fee value and freight threshold.	Management / Sales	Medium
6	Define PDF disclaimer text and document footer content.	Management	Medium

7	Confirm hosting platform and deployment environment.	Technical Team Leader	High
8	Define the access request mechanism for merchant onboarding (email or web form).	Technical Team Leader	Low
9	Confirm whether drawing files are emailed as attachments or accessed via system link.	Technical Team Leader	Low
10	Confirm initial set of merchant organisations to be onboarded at launch.	Sales / Operations	Medium

13. Glossary

Term	Definition
FIS	Free Into Store — delivery included at no additional charge
isMerchant	A flag on an organisation record that controls access to pricing and estimation functionality
Job	A top-level project record in the system representing a construction project
LVL	Laminated Veneer Lumber — an engineered timber product
Member Schedule	A background list of timber members and quantities derived from estimate inputs. Stored internally, not shown to users.
Organisation	A merchant or pre-nailer business entity in the system. Contains one or more users.
Pre-nailer	A manufacturer that pre-fabricates framed wall and floor systems using structural timber
Price Book	A set of per-lineal-metre wholesale prices for Lumberbank products, specific to a merchant organisation
Sub-Job	A discrete floor or roof system within a Job, with its own estimate inputs and sub-total
Wholesale Cost	The price charged by Lumberbank to the merchant, exclusive of GST. Not a retail or end-client price.

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